

Thévenin Equivalents & Loading

End-of-day write-up

17 June 2026 · Summer self-study, Day 2 PM

Thévenin in three lines

- Any **linear two-terminal network** looks, from its terminals, like one V_{Th} **in series with one** R_{Th} .
- V_{Th} = the **open-circuit voltage** (measured with nothing connected).
- R_{Th} = the resistance **looking back in** with sources killed: a **voltage source** $\rightarrow$ **short**, a **current source** $\rightarrow$ **open**.

Worked example: a loaded voltage divider

Problem. A $10\text{ k}\Omega$ / $10\text{ k}\Omega$ divider runs from 5 V . **(a)** Find its Thévenin equivalent. **(b)** Find the output voltage when loaded by $10\text{ k}\Omega$, $1\text{ k}\Omega$ and $100\text{ }\Omega$.

Assumptions & laws. Ideal 5 V source; resistor values exact; voltage-divider rule and Thévenin reduction; meter assumed not to load the circuit.

(a) The Thévenin equivalent

Open-circuit voltage (no load $\Rightarrow$ no current through the divider, clean ratio):

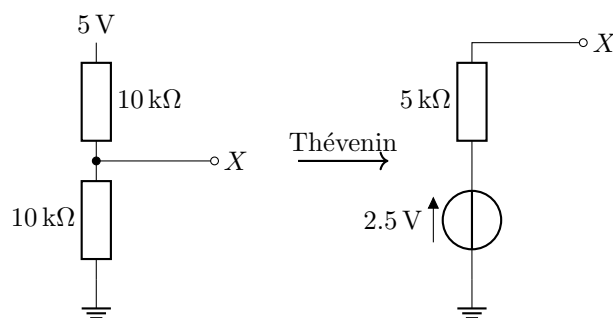
$$V_{Th} = 5 \times \frac{10\text{ k}}{10\text{ k} + 10\text{ k}} = 2.5\text{ V}$$

Look-back resistance — kill the source (short the 5 V); the two 10 k now hang from the output to ground in **parallel**:

$$R_{Th} = 10\text{ k} \parallel 10\text{ k} = \frac{10\text{ k} \times 10\text{ k}}{10\text{ k} + 10\text{ k}} = 5\text{ k}\Omega$$

$$V_{Th} = 2.5\text{ V}$$

$$R_{Th} = 5\text{ k}\Omega$$



(b) Output under load

The load R_L and R_{Th} form a fresh divider, so:

$$V_{out} = V_{Th} \frac{R_L}{R_{Th} + R_L} = 2.5 \frac{R_L}{5\text{ k} + R_L}$$

- $R_L = 10\text{ k}\Omega$: $2.5 \times \frac{10}{5 + 10} = \mathbf{1.67\text{ V}}$
- $R_L = 1\text{ k}\Omega$: $2.5 \times \frac{1}{5 + 1} = \mathbf{0.42\text{ V}}$

- $R_L = 100\ \Omega$: $2.5 \times \frac{0.1}{5 + 0.1} = \mathbf{0.049\ V}$

Key insight. The output is **not** a stiff 2.5 V. The load pulls current through R_{Th} , dropping voltage across it, so less reaches the output. The **smaller the load relative to R_{Th} , the bigger the drop** — a $100\ \Omega$ load ($\ll 5\ \text{k}$) almost collapses it to zero. Rule of thumb: keep the load $\geq 10 \times R_{Th}$ to stay within $\sim 10\%$.

What the labs taught — the keepers

Meter habits

- **Voltage** $\rightarrow$ meter in **parallel** (across the part). **Current** $\rightarrow$ meter in **series** (in the loop, mA socket). Never an ammeter across the supply. **Resistance** $\rightarrow$ only on an **isolated, unpowered** part.
- Scope ground clips are all **common and earthed** $\rightarrow$ use **one ground**. For a part not touching ground, read both ends to ground and **subtract** (Math A–B). A channel reads (**tip**) – (**its own ground**).
- **Every meter loads** the circuit (its R_{in} vs R_{Th}): high R_{in} = negligible, low R_{in} = big drop. The **10 $\times$ rule**.

Core laws

- **KVL**: drops round a loop add up to the supply ($V_R + V_{LED} = V_s$).
- **Ohm's law** $V = IR$ — pulled from your own current sweep.
- **Power** $P = VI = I^2R = V^2/R$; every part has a **wattage rating** — exceed it $\rightarrow$ heat $\rightarrow$ failure. A **fuse** is a sacrificial link that blows above its rated current.

The LED

- It **clamps its voltage at $\sim 2\ \text{V}$** and does **not limit its own current**. What destroys it is **current** (heat), not voltage.
- So it needs a **series resistor**: the resistor takes the leftover voltage ($V_R = V_{wiper} - 2$) and **sets the current**, $I = (V_{wiper} - 2)/470$.

Thévenin & loading

- Any source/divider at two terminals = V_{Th} **in series with** R_{Th} (open-circuit voltage; resistance with sources killed).
- **Loading drops the output**: $V = V_{Th} R_L / (R_{Th} + R_L)$ — watched 2.5 V fall to 0.049 V.
- A **pot is an adjustable divider**: one track split by the wiper into $R_{top} + R_{bot}$ (always = full value); the wiper voltage is the output, sliding $0 \rightarrow 5\ \text{V}$.

Real sources

- A real battery/PSU = **ideal source + internal resistance** (a Thévenin source). Shorting gives a **finite** current ($V/R_{internal}$); a current-limited PSU caps it.